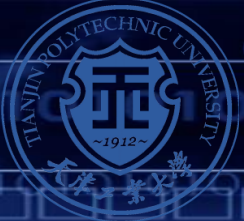




实验三 防火墙安全策略配置



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实验目的

2

实验拓扑设计与搭建

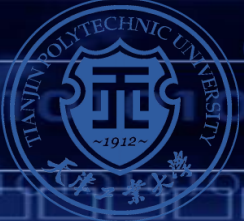
3

实验网络配置



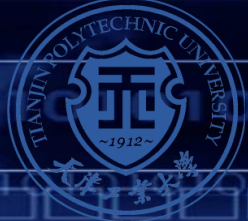
1

实验目的



实验目的

- ❖ 掌握防火墙安全策略配置流程
- ❖ 掌握对象文件在安全策略中的应用方法
- ❖ 掌握防火墙安全策略的配置命令



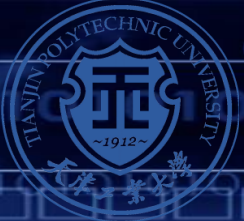
实验要求

- ❖ 某企业部署两台业务服务器，其中Server1通过TCP 80端口对外提供web服务
- ❖ 需要通过FW进行访问控制，
 - **禁止时段**：8：00 ~ 8：10 （根据实验具体情况定义）
 - **禁止IP地址为**：10.1.x.y、10.2 .x.y 的两台PC使用这台服务器对外提供的服务。
x: 代表学号倒数第三位； **y**:学号倒数后两位
 - **允许**：其他PC在任何时间都可以使用这两台服务器对外提供的服务。

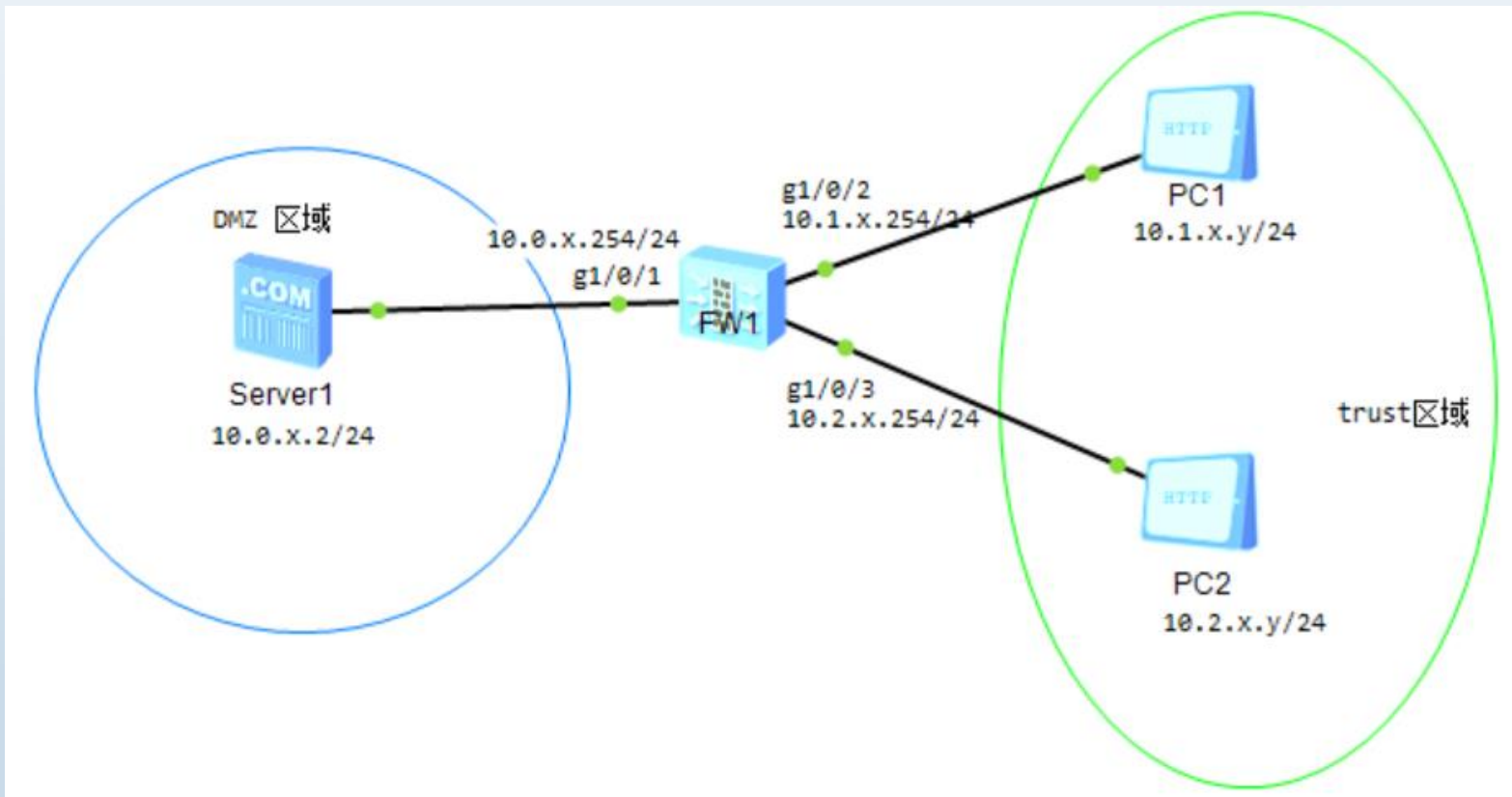


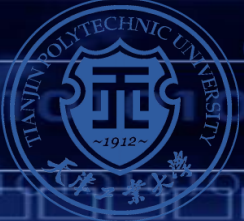
2

实验拓扑



实验拓扑结构

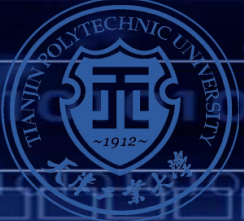




实验设备

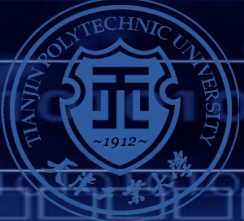
❖ 实验设备：

- 1台usg6000防火墙
- 2台客户pc机
- 1台服务器



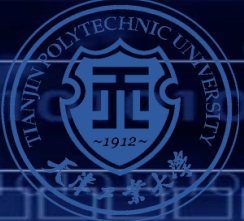
设备互联

序号	设备名称	设备接口	互联设备	互联接口	所处区域
1	FW1	g1/0/1	server1	e0/0/0	dmz
2	FW1	g1/0/2	pc1	e0/0/0	trust
3	FW1	g1/0/3	pc2	e0/0/0	trust



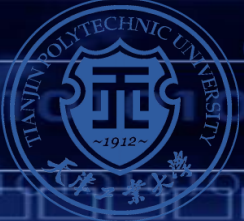
IP地址规划

序号	设备	端口号	IP地址	子网掩码	网关地址
1	FW1	G1/0/1	10.0.x.254	255.255.255.0	无
2	FW1	G1/0/2	10.1.x. 254	255.255.255.0	无
3	FW1	G1/0/3	10.2.x. 254	255.255.255.0	无
4	PC1	E0/0/0	10.1.x.y	255.255.255.0	10.1.x. 254
5	PC2	E0/0/0	10.2.x.y	255.255.255.0	10.2.x. 254
6	server1	E0/0/0	10.0.x.y	255.255.255.0	10.0.x. 254



实验步骤

- ❖ 配置设备基本信息
- ❖ 配置互联路由实现网络通信
- ❖ 配置防火墙，开启所有服务
- ❖ 建立安全策略
- ❖ 添加相关安全策略
- ❖ 设置各安全策略的行为



Server1基础配置

❖ 鼠标双击拓扑图中server1图标，弹出配置窗口

Server1

基础配置 服务器信息 日志信息

Mac地址: 54-89-98-33-0E-D0 (格式:00-01-02-03-04-05)

IPV4配置

本机地址: 10 . 0 . 0 . 2 子网掩码: 255 . 255 . 255 . 0

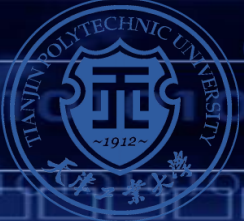
网关: 10 . 0 . 0 . 254 域名服务器: 0 . 0 . 0 . 0

PING测试

目的IPV4: 0 . 0 . 0 . 0 次数: 发送

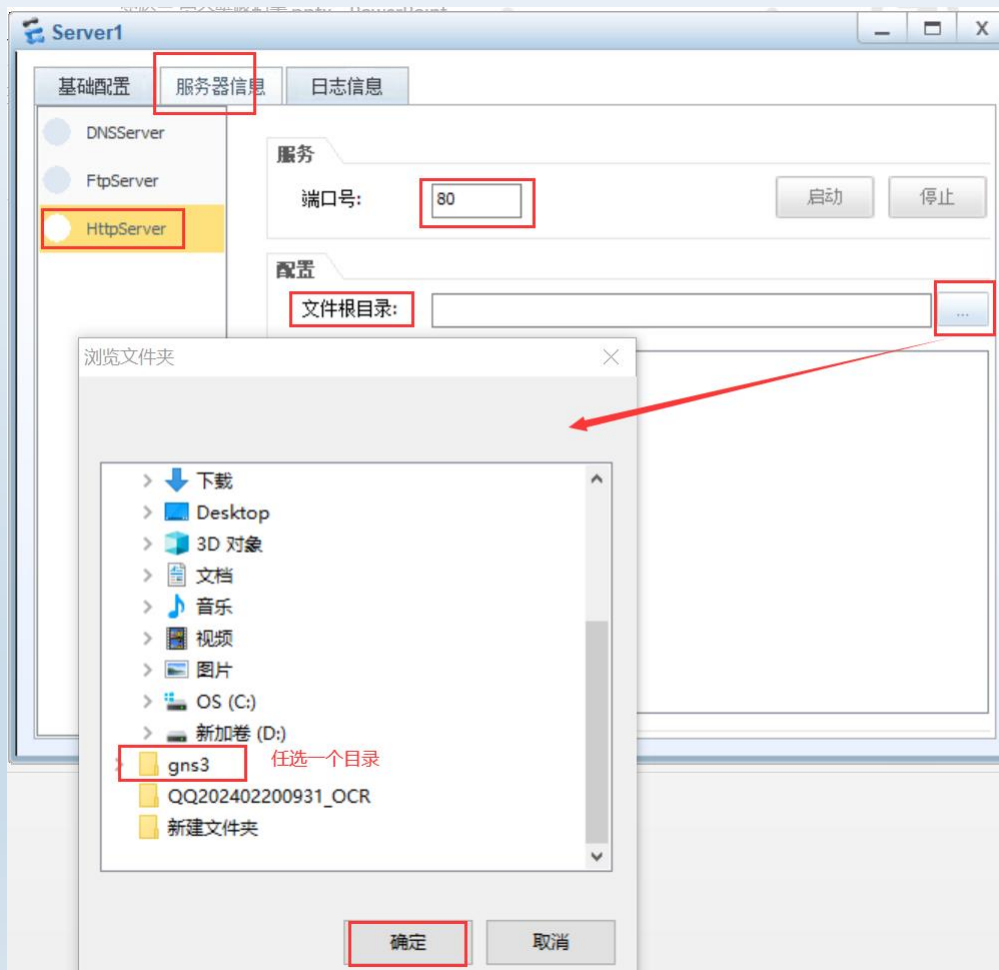
本机状态: 设备启动 ping 成功: 0 失败: 0

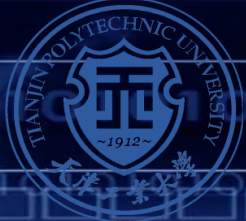
保存



Server1 基础配置

❖ 选择“服务器信息标签”，完成服务端口目录配置，点击启动





PC1基础配置要求

PC1

基础配置 客户端信息 日志信息

Mac地址: 54-89-98-F8-5A-8B (格式:00-01-02-03-04-05)

IPV4配置

本机地址: 10 . 1 . 1 . 2 子网掩码: 255 . 255 . 255 . 0

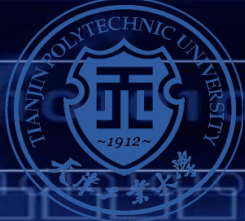
网关: 10 . 1 . 1 . 254 域名服务器: 0 . 0 . 0 . 0

PING测试

目的IPV4: 0 . 0 . 0 . 0 次数: 发送

本机状态: 设备启动 ping 成功: 0 失败: 0

保存



PC2基础配置要求

PC2

基础配置 | 客户端信息 | 日志信息

Mac地址: 54-89-98-BD-02-80 (格式:00-01-02-03-04-05)

IPv4配置

本机地址: 10 . 2 . 2 . 2 子网掩码: 255 . 255 . 255 . 0

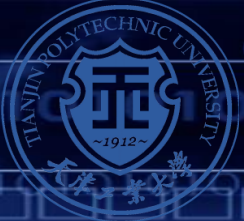
网关: 10 . 2 . 2 . 254 域名服务器: 0 . 0 . 0 . 0

PING测试

目的IPv4: 0 . 0 . 0 . 0 次数: 发送

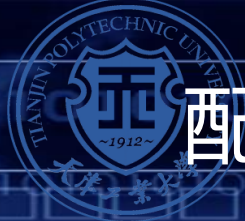
本机状态: 设备启动 ping 成功: 0 失败: 0

保存



防火墙配置步骤

- ❖ 配置接口IP地址和安全区域，完成网络基本参数配置。
- ❖ 配置名称为server_deny的地址集，将几个不允许访问服务器的IP地址加入地址集
- ❖ 配置名称为time_deny的时间段，指定PC不允许访问服务器的时间
- ❖ 为Server1配置自定义服务集server1_port，将服务器的端口加入服务集
- ❖ 配置安全策略规则，引用之前配置的地址集、时间段及服务集



配置接口G1/0/1的IP地址和安全区域

- ❖ 配置GigabitEthernet 1/0/1接口IP地址，将接口加入dmz域。

```
<FW> system-view
```

```
[FW] interface GigabitEthernet 1/0/1
```

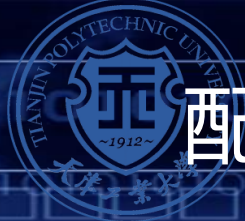
```
[FW-GigabitEthernet1/0/1] ip address 10.0.0.254 24
```

```
[FW-GigabitEthernet1/0/1] quit
```

```
[FW] firewall zone dmz
```

```
[FW-zone-dmz] add interface GigabitEthernet 1/0/1
```

```
[FW-zone-dmz] quit
```



配置接口G1/0/2的IP地址和安全区域

- ❖ 配置GigabitEthernet 1/0/2接口IP地址，将接口加入dmz域。

```
<FW> system-view
```

```
[FW] interface GigabitEthernet 1/0/2
```

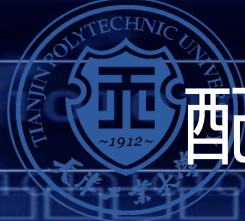
```
[FW-GigabitEthernet1/0/1] ip address 10.1.1.254 24
```

```
[FW-GigabitEthernet1/0/1] quit
```

```
[FW] firewall zone trust
```

```
[FW-zone-trust] add interface GigabitEthernet 1/0/2
```

```
[FW-zone-trust] quit
```



配置接口G1/0/3的IP地址和安全区域

- ❖ 配置GigabitEthernet 1/0/3接口IP地址，将接口加入dmz域。

```
<FW> system-view
```

```
[FW] interface GigabitEthernet 1/0/3
```

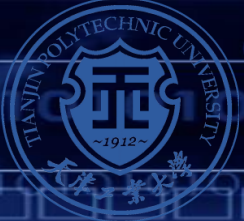
```
[FW-GigabitEthernet1/0/1] ip address 10.2.2.254 24
```

```
[FW-GigabitEthernet1/0/1] quit
```

```
[FW] firewall zone trust
```

```
[FW-zone-trust] add interface GigabitEthernet 1/0/3
```

```
[FW-zone-trust] quit
```

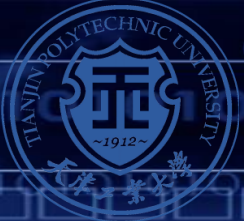


查看接口配置信息

❖ [FW] display ip interface brief

```
(E): E-Trunk down
The number of interface that is UP in Physical is 5
The number of interface that is DOWN in Physical is 5
The number of interface that is UP in Protocol is 5
The number of interface that is DOWN in Protocol is 5
```

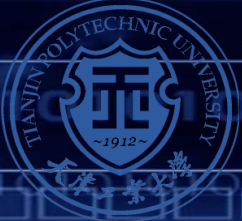
Interface	IP Address/Mask	Physical
GigabitEthernet0/0/0	192.168.0.1/24	down
GigabitEthernet1/0/0	unassigned	down
GigabitEthernet1/0/1	10.0.0.254/24	up
GigabitEthernet1/0/2	10.1.1.254/24	up
GigabitEthernet1/0/3	10.2.2.254/24	up
GigabitEthernet1/0/4	unassigned	down
GigabitEthernet1/0/5	unassigned	down
GigabitEthernet1/0/6	unassigned	down
NULL0	unassigned	up
Virtual-if0	unassigned	up



修改缺省安全策略

❖ 配置缺省安全策略

- security-policy
- default action permit



中间结果验证-连通性验证

Client1

基础配置

客户端信息

日志信息

Mac地址:54-89-98-37-23-A7(格式:00-01-02-03-04-05)

IPv4配置

本机地址:2 . 2 . 2 . 2子网掩码:255 . 255 . 255 . 0

网关:2 . 2 . 2 . 254域名服务器:0 . 0 . 0 . 0

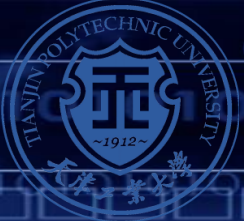
PING测试

目的IPv4:1 . 1 . 1 . 1次数:5发送

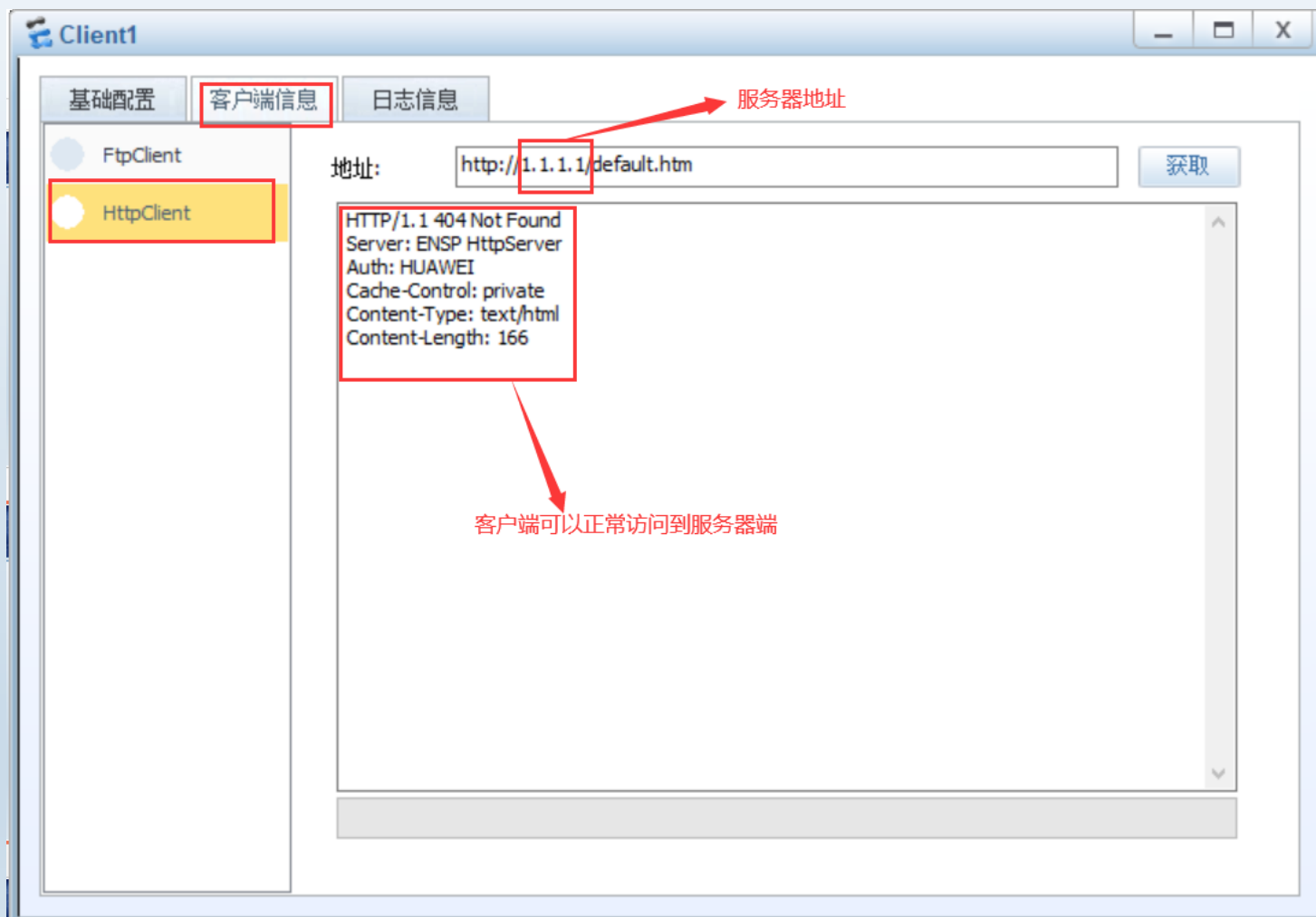
服务器地址发送包的个数

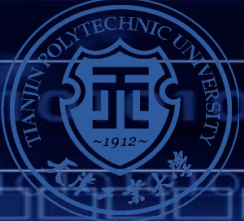
本机状态:设备启动ping 成功: 5 失败: 0

保存



中间结果验证-web服务验证





配置地址集

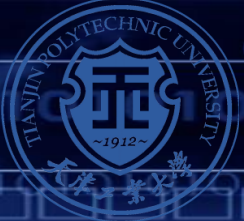
- ❖ 配置名称为server_deny的地址集，将几个不允许访问服务器的IP地址加入地址集。

```
[FW] ip address-set server_deny type object
```

```
[FW-object-address-set-server_deny] address 10.1.1.2 mask 32
```

```
[FW-object-address-set-server_deny] address 10.2.2.2 mask 32
```

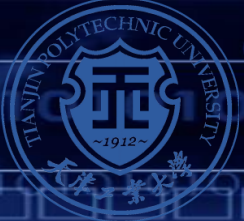
```
[FW-object-address-set-server_deny] quit
```



查看地址集配置情况

[FW]display ip address-set type object

```
[FW]disp ip address-set type object
2024-03-07 12:39:33.710
Object address-set maximum number(s): 256
Object address-set total number(s): 1
Object address-set item total number(s): 2
Object address-set reference total number(s): 0
Address-set: server_deny
Type: object
Item number(s): 2
Reference number(s): 0
Item(s):
address 0 10.1.1.2 mask 32
address 1 10.2.2.2 mask 32
```



配置时间段表

❖ 配置名称为time_deny的时间段，指定PC不允许访问服务器的时间。

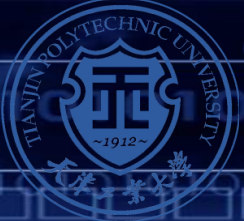
[FW] display clock 查看系统时间

[FW] time-range time_deny

[FW-time-range-time_deny] period-range 08:30:00 to 09:00:00
mon tue wed thu fri sat sun

(时间段可以结合实验时间自行调整)

[FW-time-range-time_deny] quit

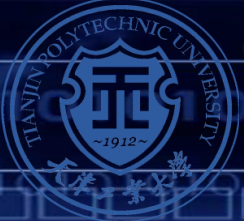


查看时间段配置

❖ [FW] display time-range time_deny

```
[FW]display time-range time_deny
2024-03-07 12:50:02.140
Current time is 12:50:02 3-7-2024 Thursday

Time-range : time deny ( Inactive )
period-range 20:45:00 to 21:00:00 daily
```



为Server1配置自定义服务集

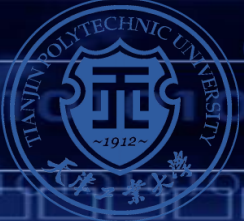
- ❖ 为Server1配置自定义服务集server1_port，将服务器的端口加入服务集。

```
[FW] ip service-set server1_port type object
```

```
[FW-object-service-set-server1_port]
```

```
service protocol TCP source-port 0 to 65535 destination-port 80
```

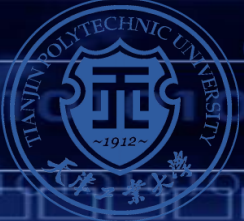
```
[FW-object-service-set-server1_port] quit
```



查看server1自定义服务集

❖ [FW]display ip service-set type object

```
[FW]display ip service-set type object
2024-03-07 12:54:56.030
Object service-set maximum number(s): 256
Object service-set total number(s): 1
Object service-set item total number(s): 1
Object service-set reference total number(s): 0
Service-set Name: server1_port
Type: object
Item number(s): 1
Reference number(s): 0
Item(s):
service 0 protocol tcp source-port 0 to 65535 destination-port 80
```



配置安全策略规则

❖ 配置安全策略规则，引用之前配置的地址集、时间段及服务集。

❖ # 限制PC使用Server1对外提供的服务的安全策略

```
[FW] security-policy
```

```
[FW-policy-security] rule name policy_sec_deny1
```

```
[FW-policy-security-rule-policy_sec_deny1] source-zone trust
```

```
[FW-policy-security-rule-policy_sec_deny1] destination-zone dmz
```

```
[FW-policy-security-rule-policy_sec_deny1]
```

```
source-address address-set server_deny
```

```
[FW-policy-security-rule-policy_sec_deny1]
```

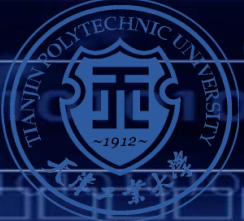
```
destination-address 10.0.0.2 32
```

```
[FW-policy-security-rule-policy_sec_deny1] service server1_port
```

```
[FW-policy-security-rule-policy_sec_deny1] time-range time_deny
```

```
[FW-policy-security-rule-policy_sec_deny1] action deny
```

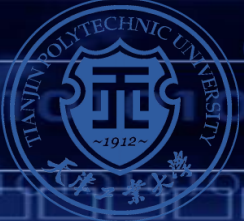
```
[FW-policy-security-rule-policy_sec_deny1] quit
```



查看安全策略配置

❖ [FW] display security-policy rule name policy_sec_deny1

```
[FW]display security-policy rule name policy_sec_deny1
2024-03-07 13:01:48.760
(0 times matched)
rule name policy_sec_deny1
  source-zone trust
  destination-zone dmz
  source-address address-set server_deny
  destination-address 10.0.0.2 mask 255.255.255.255
  service server1_port
  time-range time_deny
  action deny
```



配置安全策略规则

❖ # 允许PC使用Server1对外提供的服务的安全策略

[FW-policy-security] rule name policy_sec_permit2

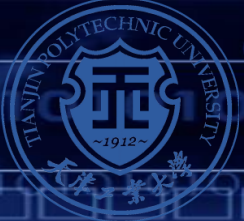
[FW-policy-security-rule-policy_sec_permit3] source-zone trust

[FW-policy-security-rule-policy_sec_permit3] destination-zone dmz

[FW-policy-security-rule-policy_sec_permit3] service server1_port

[FW-policy-security-rule-policy_sec_permit3] action permit

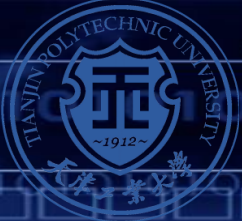
[FW-policy-security-rule-policy_sec_permit3] quit



查看安全策略配置

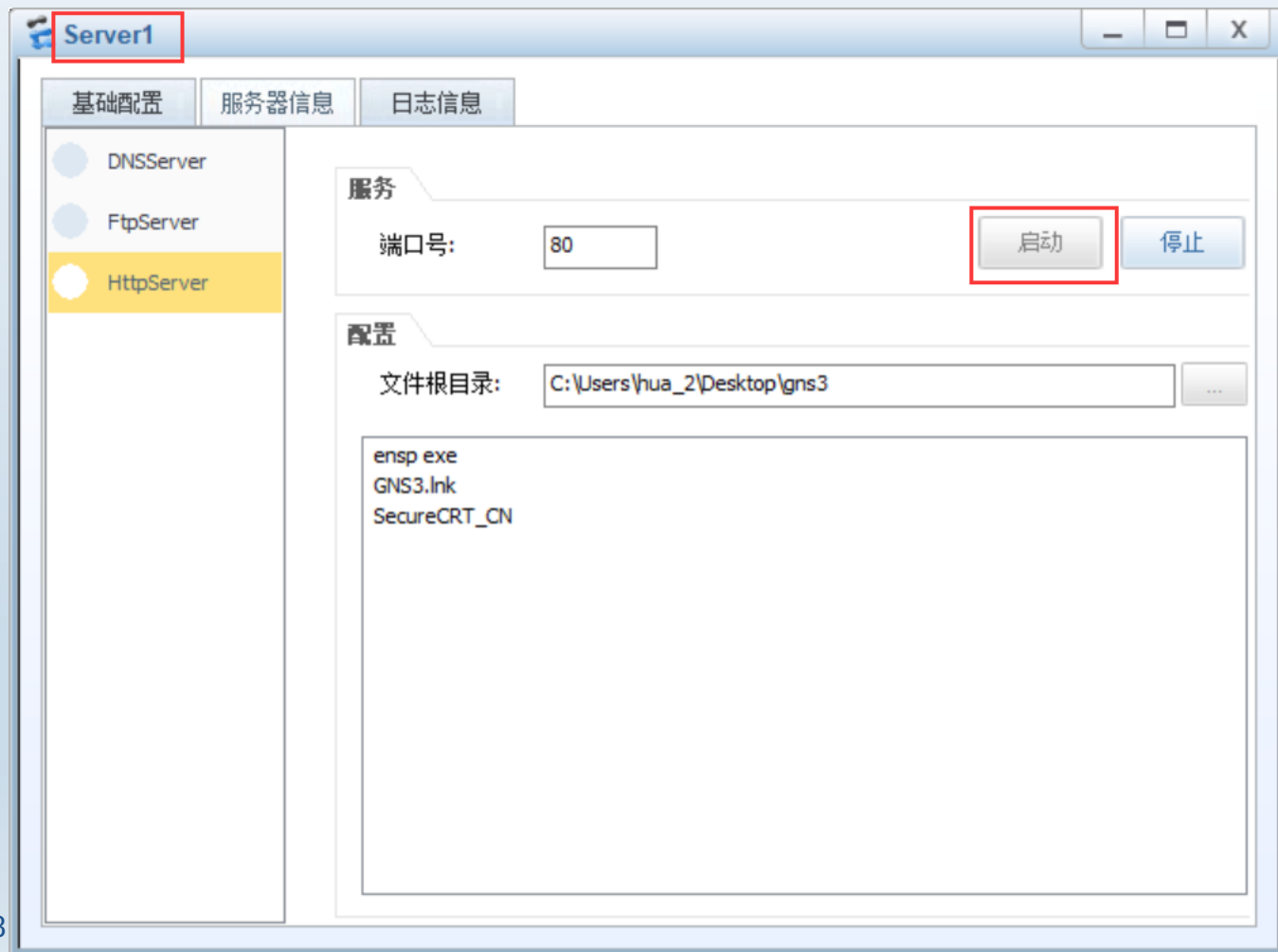
❖ [FW] display security-policy rule name policy_sec_permit2

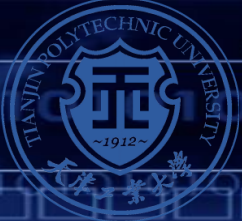
```
[FW]display security-policy rule name policy_sec_permit2
2024-03-07 13:08:43.910
(0 times matched)
rule name policy sec permit2
source-zone trust
destination-zone dmz
service server1_port
action permit
```



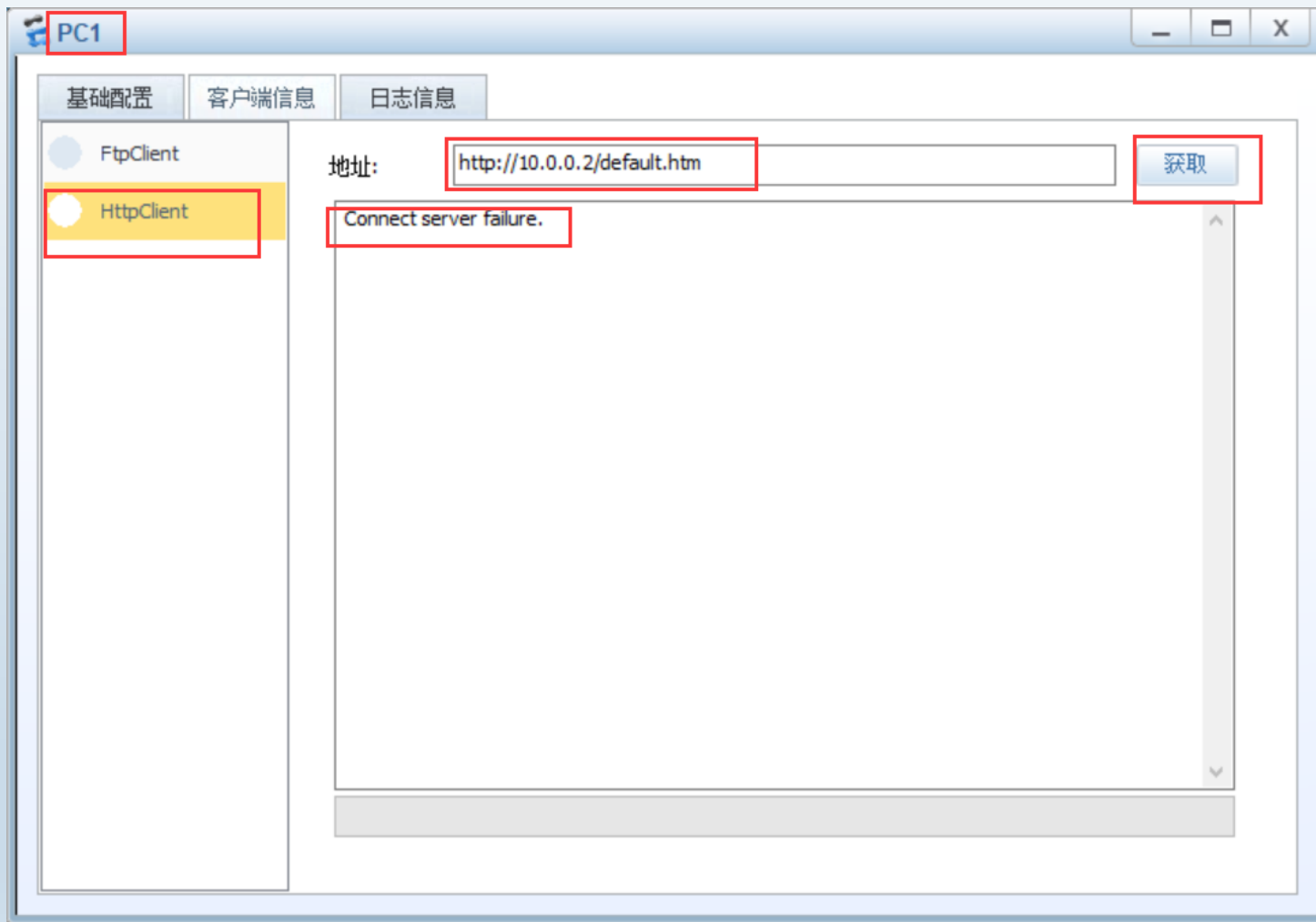
结果验证

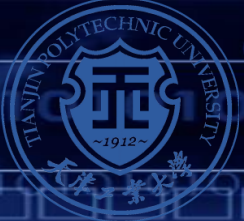
❖ 启动server1





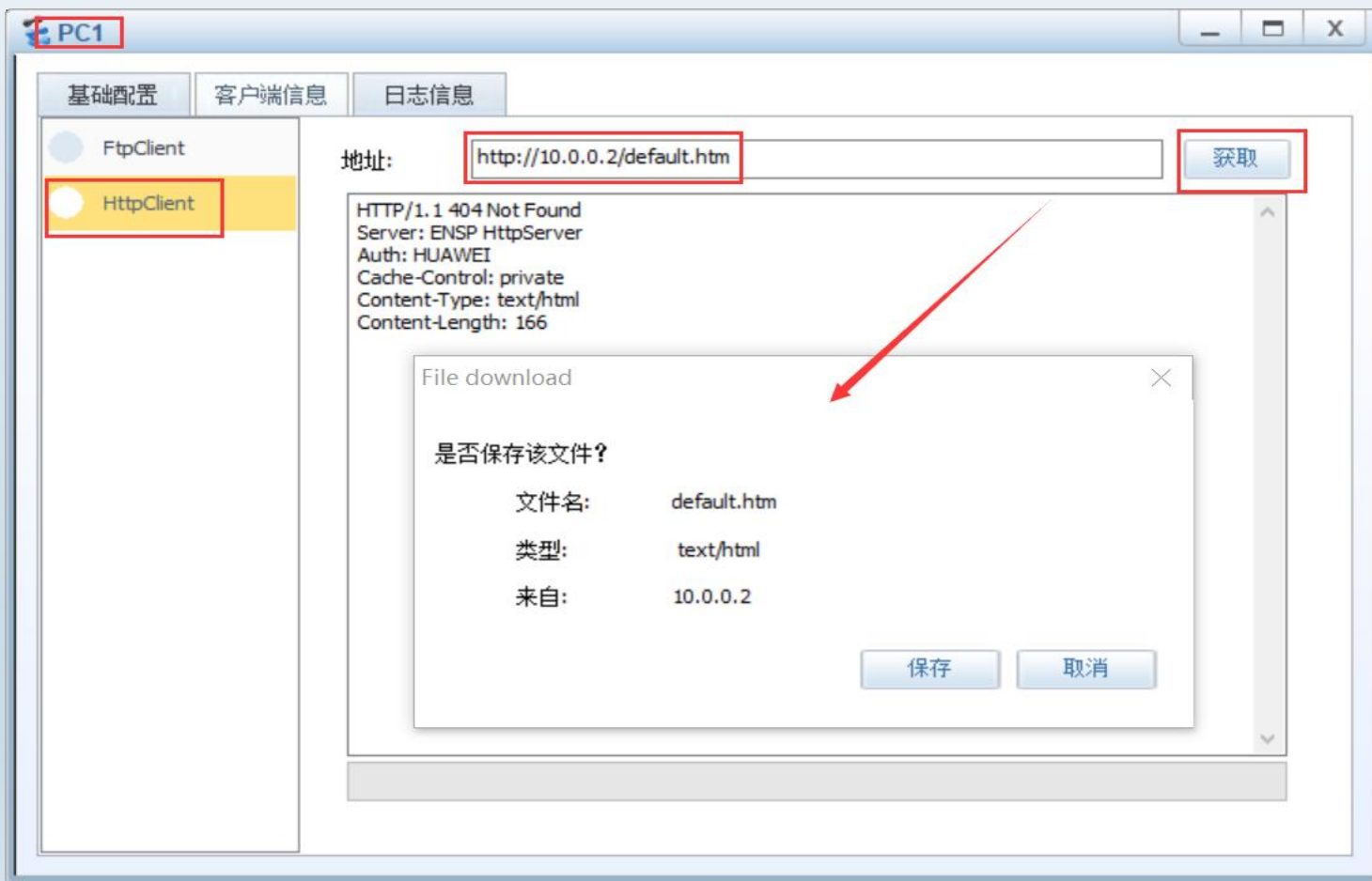
规定时间内无法访问服务器





过了时间段后

❖ pc1可以访问server1正常





谢谢!